

Module 1 – Python Basics

1. Python Introduction

What is Python?

Python is a high-level, easy-to-learn programming language widely used in data science, web development, and automation.

Why Python?

- Simple and readable syntax
 - Large community support
 - Rich libraries for data science
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2. Data Types

What are Data Types?

Data types define the type of data a variable can store.

Common Data Types

- int → Integer numbers
- float → Decimal numbers
- str → Text
- list → Collection of items
- dict → Key-value pairs

Example:

```
name = "Ali"  
age = 25  
marks = [90, 85, 88]
```

3. Control Flow

What is Control Flow?

Control flow determines how a program executes based on conditions and loops.

Types

- if-else statements
- for loops
- while loops

Example:

```
if age > 18:  
    print("Adult")
```

4. Functions

What are Functions?

Functions are reusable blocks of code used to perform specific tasks.

Why Use Functions

- Avoid repetition
- Improve readability
- Modular code

Example:

```
def add(a, b):  
    return a + b
```

5. Modules

What are Modules?

Modules are files that contain reusable Python code.

Why Modules

- Organize code
- Reuse functionality
- Improve maintainability

Example:

```
import math  
  
print(math.sqrt(16))
```

6. Key Concepts to Remember

- Python is simple and powerful
- Data types define stored values
- Control flow manages execution
- Functions reuse logic
- Modules organize code

7. Summary

Python basics form the foundation of data science. Understanding variables, data types, control flow, functions, and modules is essential before working with data analysis and machine learning.